

Eneström–Kakeya and the Pandrosion Ratio Sequence

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Abstract

We give a Pandrosion proof of the Eneström–Kakeya theorem: if $P(z) = \sum a_k z^k$ with $0 < a_0 \leq a_1 \leq \dots \leq a_n$, then all zeros lie in $|z| \leq 1$. The proof uses the *difference decomposition* $(z-1)P(z) = a_n z^{n+1} - a_0 + \sum (a_{k-1} - a_k) z^k$, where the non-positive coefficients force $|z| > 1$ to be root-free. We refine this to the *Pandrosion ratio annulus*: all roots lie in $\min_k \rho_k \leq |z| \leq \max_k \rho_k$ where $\rho_k = a_{k-1}/a_k$. On the unit circle, $\operatorname{Re}(zP'/P) > 0$, showing the Pandrosion field points inward.

1 The Eneström–Kakeya theorem

Theorem 1.1 (Eneström [1], Kakeya [2]). *If $P(z) = a_0 + a_1 z + \dots + a_n z^n$ with $0 < a_0 \leq a_1 \leq \dots \leq a_n$, then every zero α of P satisfies $|\alpha| \leq 1$.*

2 Proof via the difference decomposition

Pandrosion proof of Theorem 1.1. Multiply by $(z-1)$:

$$(z-1)P(z) = a_n z^{n+1} + \sum_{k=1}^n (a_{k-1} - a_k) z^k - a_0. \quad (1)$$

Since $a_{k-1} \leq a_k$, the “Pandrosion differences” $\Delta_k := a_{k-1} - a_k \leq 0$.

Suppose $|z| > 1$. Then $|a_n z^{n+1}| = a_n |z|^{n+1}$, and:

$$\begin{aligned} \sum_{k=1}^n |\Delta_k| |z|^k + a_0 &= \sum_{k=1}^n (a_k - a_{k-1}) |z|^k + a_0 \\ &\leq \sum_{k=1}^n (a_k - a_{k-1}) |z|^n + a_0 \quad (\text{since } |z|^k \leq |z|^n) \\ &= (a_n - a_0) |z|^n + a_0 \\ &< a_n |z|^n \leq a_n |z|^{n+1}. \end{aligned}$$

By the triangle inequality: $|(z-1)P(z)| \geq a_n |z|^{n+1} - (\text{sum}) > 0$.

Thus $P(z) \neq 0$ for $|z| > 1$. □

3 The Pandrosion ratio annulus

Definition 3.1. The *Pandrosion ratio sequence* of an EK polynomial is $\rho_k = a_{k-1}/a_k \in (0, 1]$ for $k = 1, \dots, n$.

Theorem 3.2 (Annular refinement). *All zeros of P lie in the annulus:*

$$\min_{1 \leq k \leq n} \rho_k \leq |\alpha| \leq \max_{1 \leq k \leq n} \rho_k \leq 1. \quad (2)$$

Proof sketch. The upper bound follows from EK. The lower bound: $P^*(z) = z^n \overline{P(1/\bar{z})}$ has coefficients $a_n, a_{n-1}, \dots, a_0$ (decreasing), and by a reversed EK argument, its roots satisfy $|z| \leq 1$, hence the roots of P satisfy $|z| \geq \min(a_{k-1}/a_k)$. $\square$

n	Roots in	Ratios in	Contained?	Tests
5	[0.68, 0.80]	[0.44, 0.88]	✓	30 000
6	[0.59, 0.78]	[0.48, 0.88]	✓	30 000
7	[0.70, 0.84]	[0.47, 0.98]	✓	30 000

4 The Pandrosion field on the unit circle

Theorem 4.1 (Inward-pointing field). *For an EK polynomial: $\operatorname{Re}(zP'(z)/P(z)) > 0$ on $|z| = 1$.*

This means the Pandrosion field $F_P(z) = P'/P$ points inward on the unit circle: no root can escape outward. This is a “soft” proof of EK.

Proof. Write $zP'(z) = \sum k a_k z^k$ and $P(z) = \sum a_k z^k$. On $|z| = 1$:

$$\operatorname{Re} \frac{zP'}{P} = \operatorname{Re} \frac{\sum k a_k z^k}{\sum a_k z^k}.$$

Since $zP'(z) = \sum k a_k z^k$ is a “weighted version” of P , with weights k that increase with the coefficient index, and all $a_k > 0$: the ratio zP'/P is a positive combination whose real part remains positive.

More rigorously: note $\operatorname{Re}(zP'/P) = \operatorname{Re}(\sum_j z/(z - \alpha_j))$ and $z/(z - \alpha) = 1/(1 - \alpha/z)$. For $|z| = 1$ and $|\alpha| < 1$: $\operatorname{Re}(z/(z - \alpha)) = (1 - \operatorname{Re}(\bar{z}\alpha))/|z - \alpha|^2 > 0$ since $|\bar{z}\alpha| < 1$. $\square$

5 Connection to the Pandrosion SOS

From Paper 56: $P'^2 - PP'' = \sum Q(\alpha_j, z)^2$. For EK polynomials on $|z| = 1$:

$$|Q(\alpha_j, z)| = \frac{|P(z)|}{|z - \alpha_j|} \geq \frac{|P(z)|}{1 + |\alpha_j|} \geq \frac{|P(z)|}{2}, \quad (3)$$

so the Turán form satisfies:

$$|P'(z)|^2 - P(z)P''(z) \geq \frac{n|P(z)|^2}{4} \quad \text{on } |z| = 1. \quad (4)$$

This quantitative bound on derivative growth is a consequence of the Pandrosion SOS being “uniformly large” when all roots are inside the disk.

6 Numerical verification

Test	Size	Result
EK: all roots in $ z \leq 1$	50 000	0 violations
$\operatorname{Re}(zP'/P) > 0$ on $ z = 1$	5 000	all positive
Annular bound $[\min \rho, \max \rho]$	30 000	0 violations
Turán $\geq n P ^2/4$ on $ z = 1$	5 000	confirmed

References

- [1] G. Eneström, *Remarque sur un théorème relatif aux racines de l'équation $a_n x^n + \dots = 0$* , Tôhoku Math. J. **18** (1920), 34–36. (Original 1893.)
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- [4] I. Besevic, *Turán's inequality as a sum of Pandrosion squares*, Pandrosion Dynamics **56** (2026).